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Introduction: The Hardware Bottleneck

Noisy Intermediate-Scale Quantum (NISQ) Devices:

- Current quantum computers are powerful but suffer from short coherence times and noisy operations.
- If a calculation takes too long, quantum information degrades.

The VQE Solution:

- Variational Quantum Eigensolver (VQE) was developed to circumvent these severe hardware limitations.
- It is a hybrid algorithm relying on an ansatz, a parameterized quantum circuit acting as a trial wavefunction.
- Success depends entirely on balancing the expressivity of the ansatz against its circuit complexity.

The COMPASS Ansatz & The Measurement Flaw

How COMPASS Works:

- A dynamic ansatz designed to span the N-electron Hilbert space with minimal complexity.
- Uses energy sorting and operator commutativity prescreening to select impactful operators.

The Vulnerability:

- Conceptualized for an *ideal* quantum environment.
- Requires substantial pre-circuit measurements on hardware to construct the mathematical unitary.
- **The Flaw:** Hardware noise actively corrupts the testing phase, degrading the ansatz structure before computation even begins.

The COMPACT Ansatz: A First-Principle Solution

✨ Cognitive Optimization via Many-body Perturbation theory towards Ansatz Construction and Tailoring (COMPACT):

- Dynamically constructs a highly expressive ansatz through classical ab-initio Many-Body Perturbation Theory (MBPT).
- MBPT calculates complex electron interactions layer by layer from a simple baseline model.

🚀 Measurement-Free Construction:

- Requires zero pre-circuit measurements.
- Ansatz structure remains completely unaffected by quantum hardware noise.
- Operates through a hierarchical, order-by-order mathematical filtering process.

COMPACT Construction - Phase 1

Phase 1: First-Order Filtering:

- Evaluates Two-Body Cluster Operators T_2 using exact first-order perturbative amplitudes.
- Applies a strict ε_1 threshold; any operator where $|t_{ij}^{ab}| \leq \varepsilon_1$ is discarded. Surviving amplitudes form the highly significant base operator pool, $T_1^{(1)}$.

t_{ij}^{ab} is calculated as follows:

$$t_{ij}^{ab} = \frac{v_{ij}^{ab}}{\Delta_{ij}^{ab}}$$

Where,

v_{ij}^{ab} is the value of the two electron integral and,

Δ_{ij}^{ab} is the value of the orbital energy denominator.

COMPACT Construction - Phase 2

Phase 2: Second-Order Scatterers:

- Evaluates One-Body Scatterers S_μ to act as mathematical contracting agents.
- Filters against a second threshold, ϵ_2 , retaining only scatterers that pass $|s_\mu^{(1)}| > \epsilon_2$ for tensor contraction.

$s_\mu^{(1)}$ is calculated as follows:

$$s_\mu^{(1)} = \frac{v_\mu}{\Delta_\mu}$$

Where,

v_μ is the value of the one-electron transition integral and,

Δ_μ is the value of the orbital energy difference.

COMPACT Construction - Phase 3

★ Phase 3: Simulating Triples via Nested Commutators :

- Explicit three-body operators (T_3) cause a catastrophic proliferation of Pauli strings and CNOT gates.
- COMPACT searches surviving pools for pairs sharing a contractible orbital index to generate nested commutators $[\kappa_\mu, \kappa_\nu]$.

Amplitudes are calculated as follows(Simplified form):

$$Amp = \frac{v_{ij}^{am} v_{mk}^{bc}}{\Delta_{ijk}^{abc} \Delta_{mk}^{bc}}$$

🔗 The Filtering Process

- Computes the exact second-order MBPT amplitude for the resulting triple excitation.
- If the aggregate amplitude passes the final ϵ_3 threshold, the single best pair is isolated.
- Hardware natively computes the nested commutator, simulating T_3 correlation with absolute minimal parameter and gate counts.

Hamiltonian Chopping

Motivation & Approach

- Quantum simulations suffer from exponential computational scaling.
- Evaluating the Hamiltonian expectation value ($\langle \psi | H | \psi \rangle$) dominates the total simulation runtime.
- Systematically pruning low-coefficient Pauli terms (below threshold Th_1) streamlines the Hamiltonian and accelerates evaluation times.

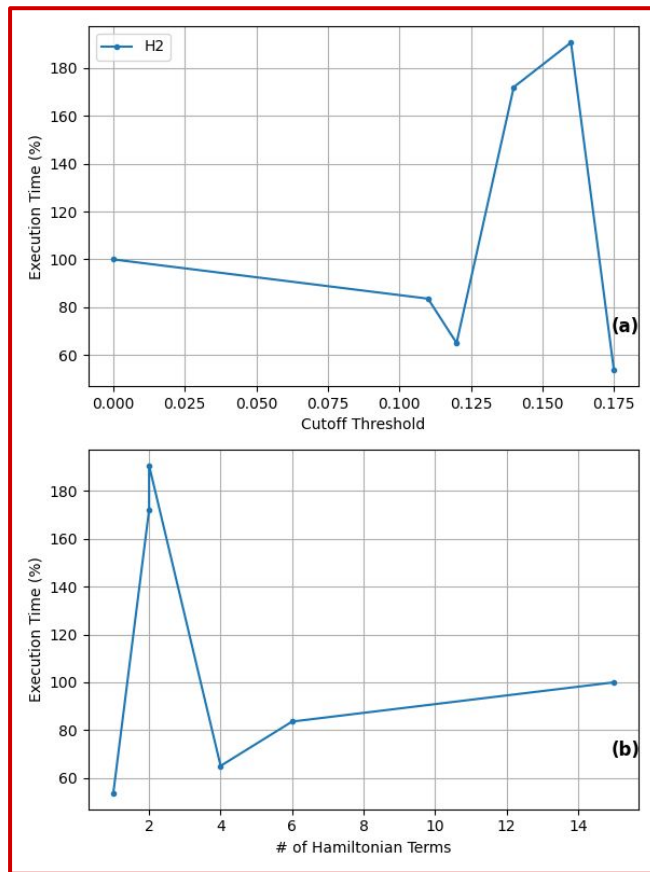
Hamiltonian Chopping

Qiskit's mapping pipeline is intercepted to discard insignificant terms prior to VQE evaluation.

```
def truncated_map(second_q_op):
    qubit_op = original_map(second_q_op)
    # Recursively prune mapped Pauli operators below the static threshold
    if isinstance(qubit_op, dict):
        return {k: SparsePauliOp(v).chop(self.ham_threshold) if v is not None else None for k, v in
qubit_op.items()}
    elif isinstance(qubit_op, list):
        return [SparsePauliOp(v).chop(self.ham_threshold) if v is not None else None for v in
qubit_op]
    elif qubit_op is not None:
        return SparsePauliOp(qubit_op).chop(self.ham_threshold)
```

Performance Analysis & Limitations

- Reducing the term count from 15 to 4 yields a clear reduction in execution time, validating the slimming strategy.
- Extreme truncation destroys the physical model, which heavily distorts the energy landscape, confuses the VQE, and causes severe execution time spikes.
- Hamiltonian truncation is highly effective, but extreme pruning on small toy models introduces mathematical instability.



Truncation Performance Metrics

Threshold (Th1)	Remaining Terms	Term Reduction	Execution Time (s)	Abs. Delta E (Ha)
0.000 (Baseline)	15	0.00%	0.5748	0.00000000
0.110	6	60.00%	0.4807	0.01479744
0.120	4	73.33%	0.3743	0.23804151
0.140	2	86.67%	0.9883	0.16401904
0.160	2	86.67%	1.0952	0.16401904
0.175	1	93.33%	0.3097	0.32417824

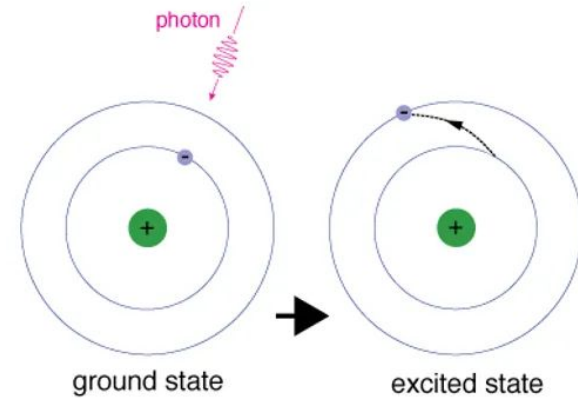
The Excited State: Energy & Transition

An **excited state** is any quantum state with energy (E_n) greater than the **ground state** (E_0). It represents a system that has captured energy.

The Mechanics:

- **Activation:** Occurs when a system absorbs external energy (light, heat, or electricity).
- **Instability:** These states are temporary; the system naturally seeks its lowest energy configuration.
- **Relaxation:** The return to ground state involves releasing energy, typically as a **photon** (light).

The energy absorbed must **exactly match** the gap between levels; there are no "in-between" states.



Computational Methods in Qiskit Nature

Calculating excited state properties requires algorithms that extend beyond the standard Ground State VQE.

Method	Algorithm Type	Working Principle	Key Strength
QEOM	Post-Processing	Applies excitation operators to a converged ground state .	Highly accurate for molecular chemistry ; very efficient.
VQD	Sequential	Finds states one-by-one using a penalty term to stay orthogonal.	Flexible; can find nth excited state without calculating others simultaneously.
SSVQE	Simultaneous	Optimizes a single circuit to map orthogonal inputs to multiple energy levels .	Faster execution; avoids the overhead of sequential "deflation" steps.

The Transition Dipole Moment

Mathematical Foundation

The Transition Dipole Moment (μ_{if}) is a vector quantity that represents the off-diagonal matrix elements of the electric dipole operator (μ) between the initial (i) and final (f) states.

Physical Significance

- **Charge Redistribution:** It quantifies the electronic displacement or polarization shift during excitation.
- **Transition Probability:** The probability of a transition is proportional to the square of the TDM magnitude ($|\mu_{if}|^2$).
- **Spatial Orientation:** Defines the transition's sensitivity to the **polarization** of the incident electromagnetic field.

Oscillator Strength (f_{S_1})

Quantifying Transition Intensity

The **oscillator strength** (f) is a dimensionless physical quantity that expresses the strength of an electronic transition relative to a classical harmonic oscillator.

$$f_{S_1} = \frac{2m_e}{3\hbar^2 e^2} \Delta E |\vec{\mu}_{S_1}|^2$$

Values near 1.0 indicate strongly allowed transitions; values $\ll 0.01$ indicate dark states or forbidden transitions.